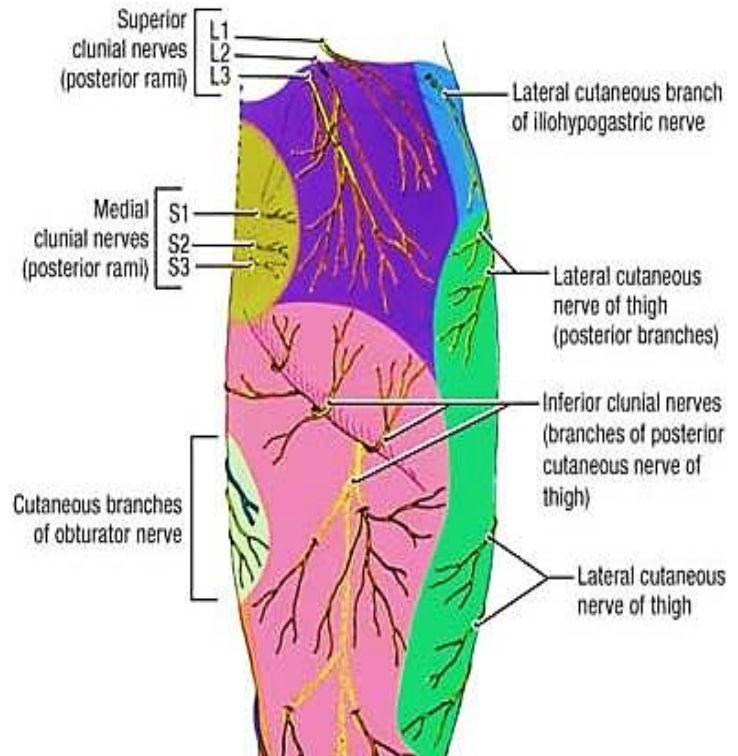


Anatomy of the gluteal region

The skin of the buttock is supplied by several small nerves derived ultimately from the region of the twelfth thoracic nerve to the third sacral nerve (T12 to S3). (Figure.1)



A bursa is found lateral to the greater trochanter, and the strong gluteal aponeurosis over the gluteus medius muscle continues inferior ward as the iliotibial tract of the fascia lata. The superolateral quadrant of the buttock is relatively free of nerves and vessels and is frequently used for intramuscular injections in order to avoid the sciatic nerve and other important structures. An alternative site is over the gluteus medius in a triangular area bounded by the anterior superior iliac spine, the tubercle of the iliac crest, and the greater trochanter (Figure.2).

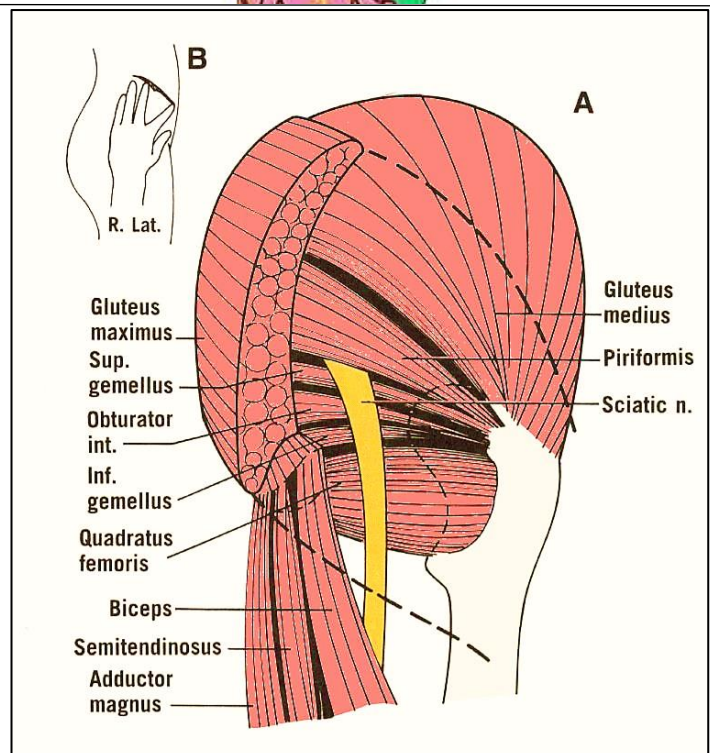


Figure 2: The gluteal region, posterior aspect.

A: The deep relations of the gluteus maximus. The sciatic nerve usually emerges inferior to the piriformis and lies on a succession of lateral rotators.

B: Intramuscular injection. An intragluteal injection may be made safely in the area between the second and third fi

Anatomy of the gluteal region

Muscles of gluteal region (Table Below)

Muscle	Origin	Insertion	Innervation	Action
Gluteus maximus	Ilium behind posterior gluteal line, dorsal surface of sacrum, & gluteal aponeurosis	Iliotibial tract & gluteal tuberosity of femur	Inferior gluteal	Extends thigh
Gluteus medius	Ilium between anterior & posterior gluteal lines	Greater trochanter of femur	Superior gluteal	Abduct & rotate thigh medially
Gluteus minimus				
Tensor fasciae latae	Anterior superior iliac spine & iliac crest	Iliotibial tract		Flexes & rotates thigh medially
Piriformis	Pelvic surface of sacrum	Greater trochanter	S.N.1 & 2	Rotate thigh laterally
Superior gemellus	Ischial spine	Obturator internus tendon	N. to obturator internus	
Obturator internus	Pelvic surface of obturator membrane & adjacent bone	Greater trochanter	Sacral plexus	
Inferior gemellus	Ischial tuberosity	Obturator internus tendon	N. to quadratus femoris	
Quadratus femoris	Ischial tuberosity	Intertrochanteric crest of femur	Sacral plexus	
Obturator externus	External surface of obturator membrane & adjacent bone	Trochanteric fossa of femur	Obturator	

The glutei maximus, medius, and minimus, from superficial to deep, form the bulk of the buttock (Figure.2). They are supplied by the gluteal nerves and vessels, which reach them through the greater sciatic foramen (Figures.3 and 4).

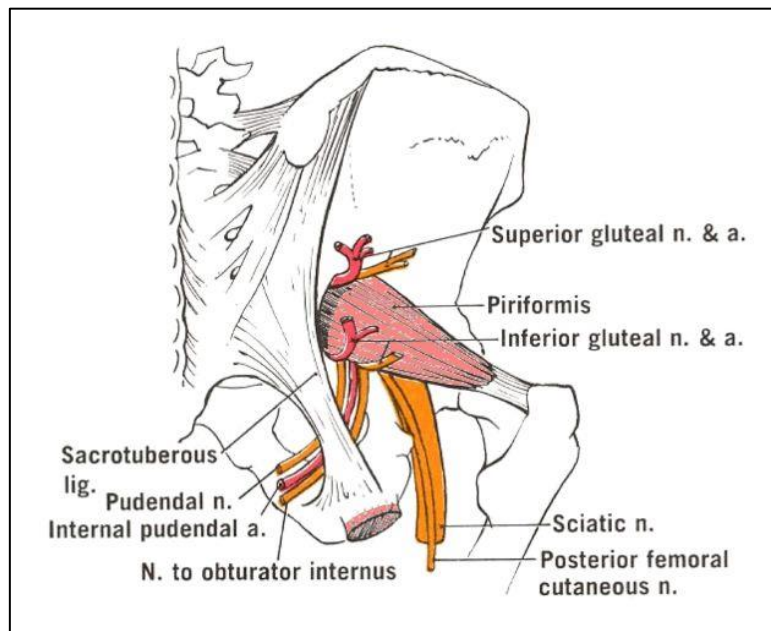


Figure.3: The arrangement of the structures emerging from the greater sciatic foramen. The foramen gives exit to the piriformis, to seven nerves (sciatic, posterior femoral cutaneous, superior gluteal, inferior gluteal, pudendal, nerve to the obturator internus, and nerve to the quadratus femoris), and to three groups of vessels (internal pudendal, superior gluteal, and inferior gluteal). The nerve to the quadratus femoris and the veins that accompany the arteries are not shown.

Anatomy of the gluteal region

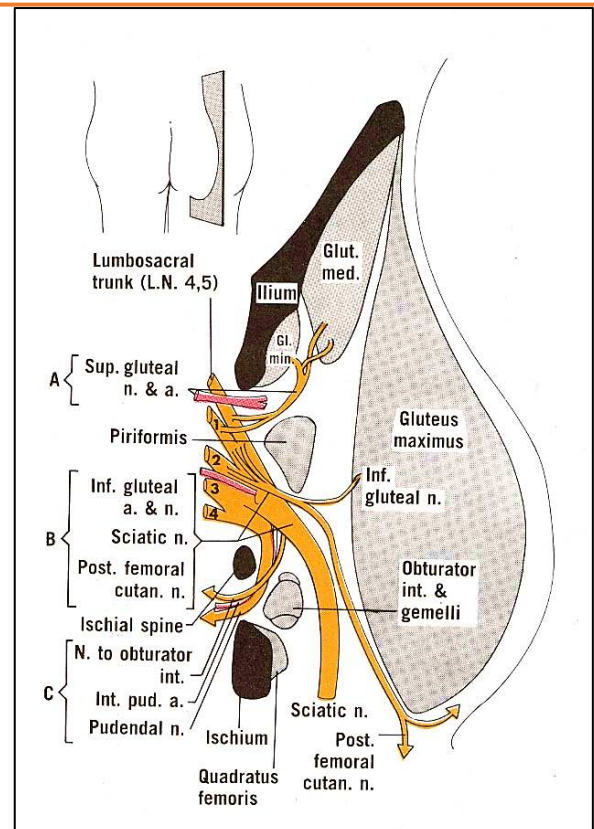
Figure.4: Oblique vertical section (see inset) to show relations in the gluteal region. Note that

(A) a nerve and artery leave the greater sciatic notch superior to the piriformis,

(B) seven nerves (the nerve to the quadratus femoris is not shown) and two arteries emerge inferior to the piriformis, and

(C) two nerves and an artery regain the pelvis through the lesser sciatic foramen. The sciatic nerve lies successively on the ischium, gemelli and obturator internus, and quadratus femoris.

(Modified from Castaing and Soutoul)



The gluteus maximus, a large muscle with numerous attachments, is a powerful extensor of the thigh or of the trunk lower limbs are in a fixed position. Surprisingly, however, it is not important posturally, is relaxed when one is standing, and is little used in walking. It is employed in running, climbing, and rising from a sitting or stooped position. It also controls flexion at the hip upon sitting down (paradoxical action). The glutei medius and minimus abduct the thigh at the hip joint and rotate it medially.

During walking, the glutei medius and minimus of the grounded limb abduct the pelvis, i.e., tilt it so that the swinging limb can clear the ground. Paralysis results in a characteristic lurching or waddling gait in which the hip on the weak side deviates laterally when that foot is in contact with the ground. A series of small lateral rotators of the thigh is found largely under cover of the gluteus maximus. One of them, the piriformis, emerges from the greater sciatic foramen, and the sciatic nerve appears at its inferior border (Figures. 2, 3 and 4).

Anatomy of the gluteal region

Nerves and vessels of gluteal region

Gluteal nerves: Several important nerves from the sacral plexus either supply or traverse the gluteal region.

- The superior gluteal nerve (L4,5, S1) (Figures. 3 and 4) passes posteriorly through the greater sciatic foramen immediately superior to the piriformis muscle. It passes anterolaterally between the gluteus medius and minimus and supplies both of those muscles as well as the tensor fasciae latae.
- The inferior gluteal nerve (L5, S1,2) (Figures. 3 and 4) traverses the greater sciatic foramen just inferior to the piriformis muscle and supplies the gluteus maximus.
- The pudendal nerve is merely passing through the gluteal region, making a brief appearance at the inferior aspect of the piriformis, only to re-enter the pelvis via the lesser sciatic foramen (Figure.4).
- The posterior femoral cutaneous nerve (S1-3) (Figures. 3 and 4) passes posteriorly through the greater sciatic foramen just inferior to the piriformis, descends deep to the gluteus maximus and continues to the middle of the posterior thigh, where it pierces the fascia, and will ultimately reach the calf.
- The largest nerve in the body, the sciatic nerve, consists of two parts, tibial and fibular, which are initially bound together and then separate at a variable level into two nerves.

A branch of the sacral plexus, the sciatic nerve (L4-S3) (Figures. 2, 3 and 4) traverses the greater sciatic foramen, below the piriformis. The fibular component, however, may pierce the piriformis (or even emerge superior to that muscle), in which case it remains separate.

The sciatic nerve descends under cover of the gluteus maximus and then along the middle of the posterior thigh. In terms of surface anatomy, the sciatic nerve leaves the pelvis a little more than a third of the way along a line from the posterior superior iliac spine to the ischial tuberosity (Figure.5). It then descends about halfway between the ischial tuberosity and the greater trochanter. The words sciatic and ischium are related.

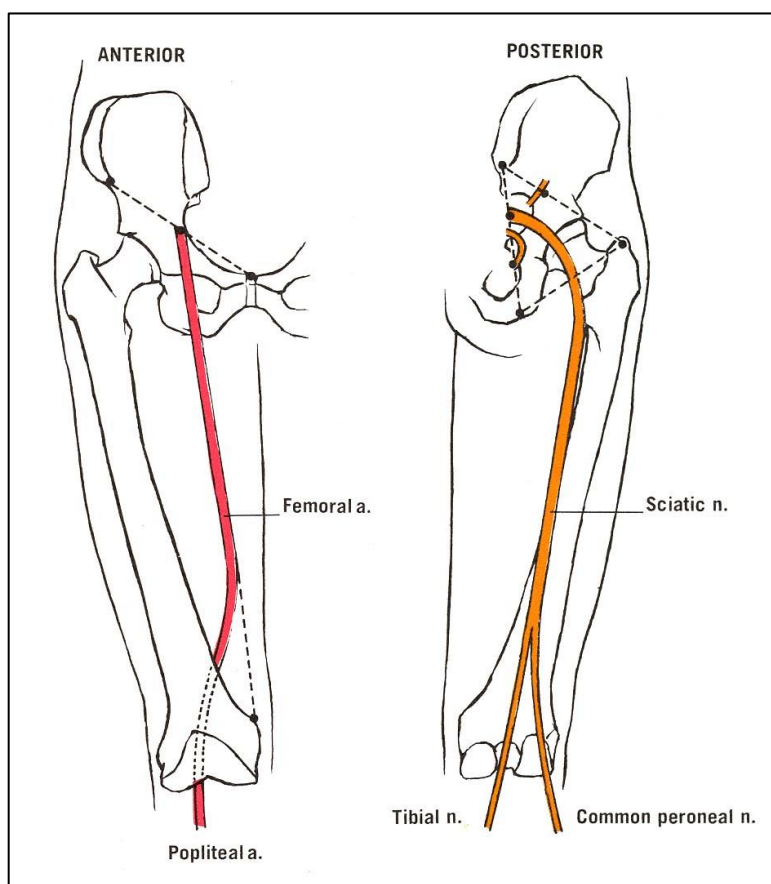


Figure.5: The surface anatomy of the femoral artery, gluteal nerves, and sciatic nerve.

Anatomy of the gluteal region

The gluteal arteries.

The gluteal arteries (Figures 6 and 3) arise, directly or indirectly, from the internal iliac artery.

- The superior gluteal artery, the largest branch of the internal iliac artery, accompanies the corresponding nerve and, under cover of the gluteus maximus, divides into branches that supply the gluteal muscles.
- The inferior gluteal artery, another branch of the internal iliac artery, accompanies the corresponding nerve, and, under cover of the gluteus maximus, lies medial to the sciatic nerve, to which it gives a companion artery.

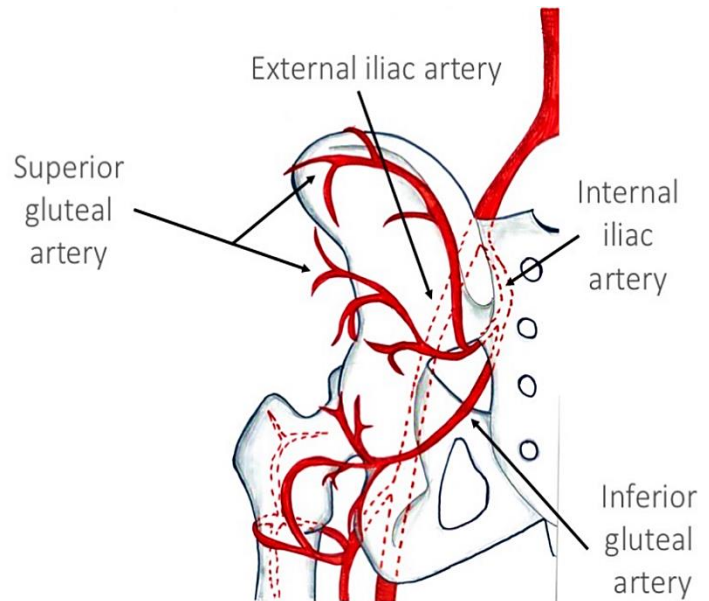


Figure.6: The Gluteal arteries network.

The gluteal veins accompany the arteries and drain into the internal iliac vein. They can return the blood from the lower limb even when the femoral vein is ligated.

The exit of the superior gluteal nerve and artery from the pelvis is indicated by the superior point of trisection of a line from the posterior superior iliac spine to the upper end of the greater trochanter (Figure.5). The exit of the inferior gluteal nerve and artery is indicated by the inferior point of trisection of a line from the posterior superior iliac spine to the ischial tuberosity.